

Internet of Things

By

Minhaz Uddin Ahmed, PhD

Department of Computer Engineering

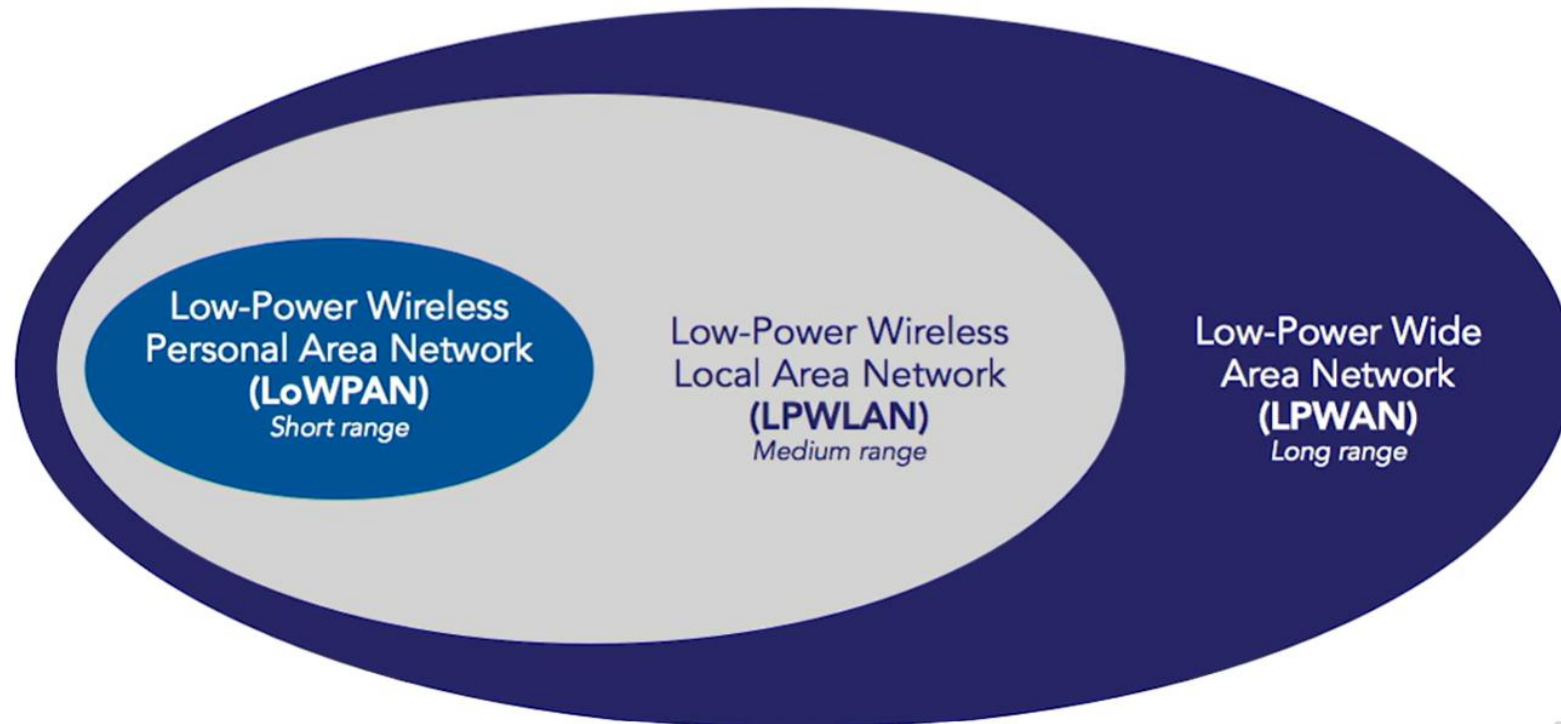
Inha University in Tashkent.

Email: minhaz.ahmed@gmail.com

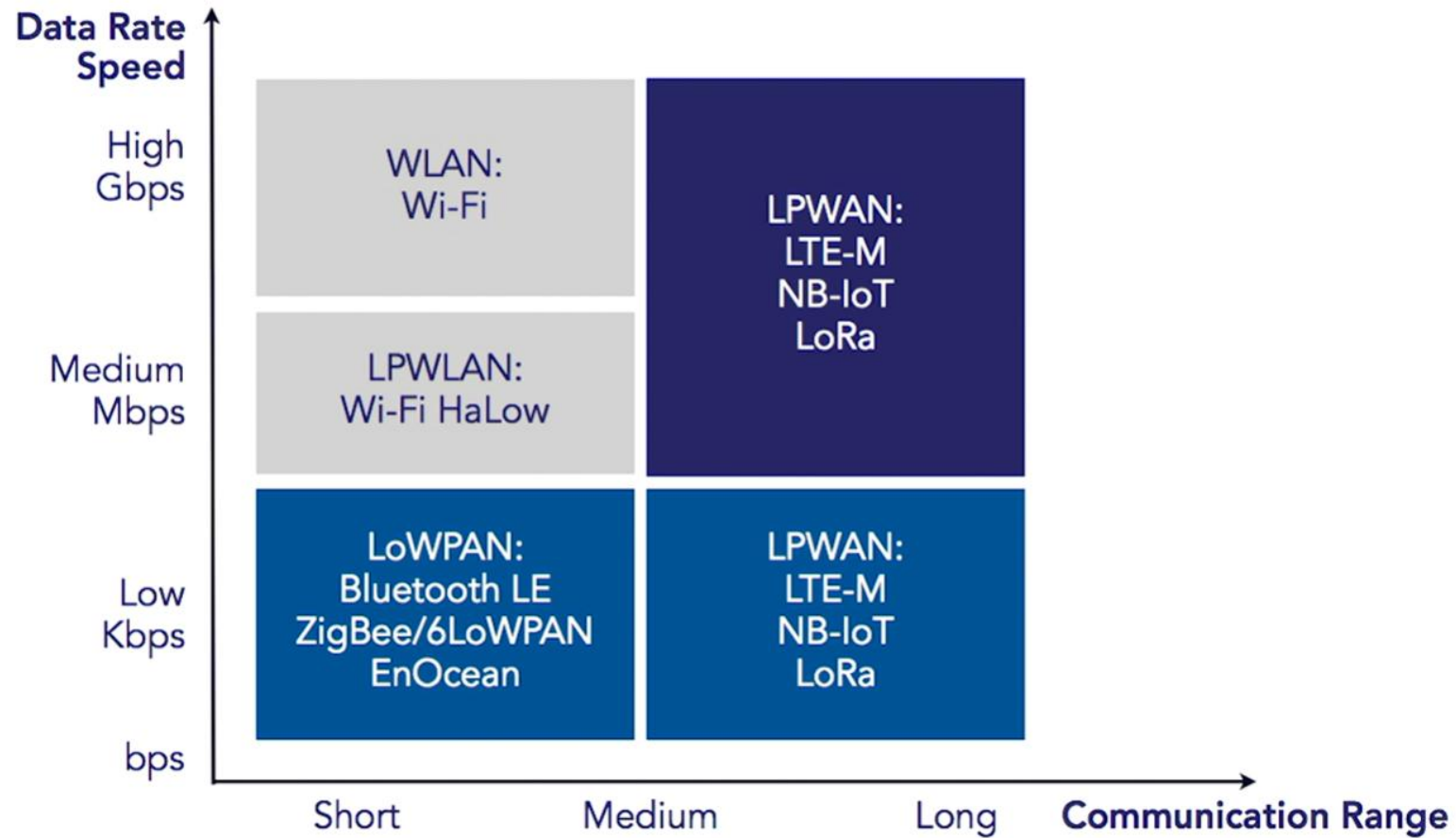
Outline

- ▶ Introduction
- ▶ IoT Foundations: Low-Power Wireless Networking
- ▶ Participatory Sensing, Industrial IoT and Automotive IoT
- ▶ Client server architecture
- ▶ Sensor Data Communication protocols
- ▶ Radio Frequency Identification Technology
- ▶ OSI reference model

Low-Power Wireless Network Types



Low-Power Wireless Networking



Examples of LoWPAN Technologies

- Bluetooth Low Energy
- ZigBee
- 6LoWPAN
- WirelessHART

Examples of LPWAN Technologies

- LTE-M (Long term evolution for machine type communication)
- NB-IoT (Narrowband IoT) supports both 2G and 4G
- EC-GSM-IoT (Extended coverage GSM IoT)
- LoRa
- SIGFOX

LoRa (Long Range) is a standard-based LPWAN technology developed by the LoRa Alliance. It also operates in sub-GHz bands, providing long-range connectivity with low power consumption. LoRa's architecture is based on a decentralized and open approach, where multiple gateways receive and relay data to the cloud.

SigFox's architecture is centralized, with all data sent to the cloud via a single base station. This centralized approach simplifies network management but may lead to potential scalability challenges in highly dense IoT deployments. Additionally, SigFox has limited uplink and downlink data rates, making it suitable for low-bandwidth applications like tracking and simple sensor monitoring.

LPWAN Technologies Uses

- ▶ Wearable devices, home appliance, to smart cities.
- ▶ From communication and network perspective IoT includes the machine to human communication.

LPWAN Technologies Uses

- Machine-to-human (M2H) communication
- Machine-to-machine (M2M) communication
- Can enable M2M and M2H communication needs with low-power wireless networking

Example of Application Domains

- Smart Home
- Smart City
- Smart manufacturing
- Smart Agriculture

Smart Agriculture

- ▶ In agriculture, 6LoWPAN can be used to deploy wireless sensor networks that monitor soil moisture, temperature, and other environmental conditions.
- ▶ This data can be used to optimize irrigation, fertilization, and other farming practices, improving efficiency and reducing waste.



Smart house system

Smart lights are equipped with sensors

- Smart lights
- Smart metering
- Access control
- Security
- And more



Smart City

Smart City

- Environmental monitoring
- Smart street lights
- Smart buildings
- And more



6LoWPAN enables each light to have an IP address, and therefore to be controlled and monitored remotely. This allows for dimming lights when not needed, monitoring for outages, and gathering data on usage.

Smart Manufacturing

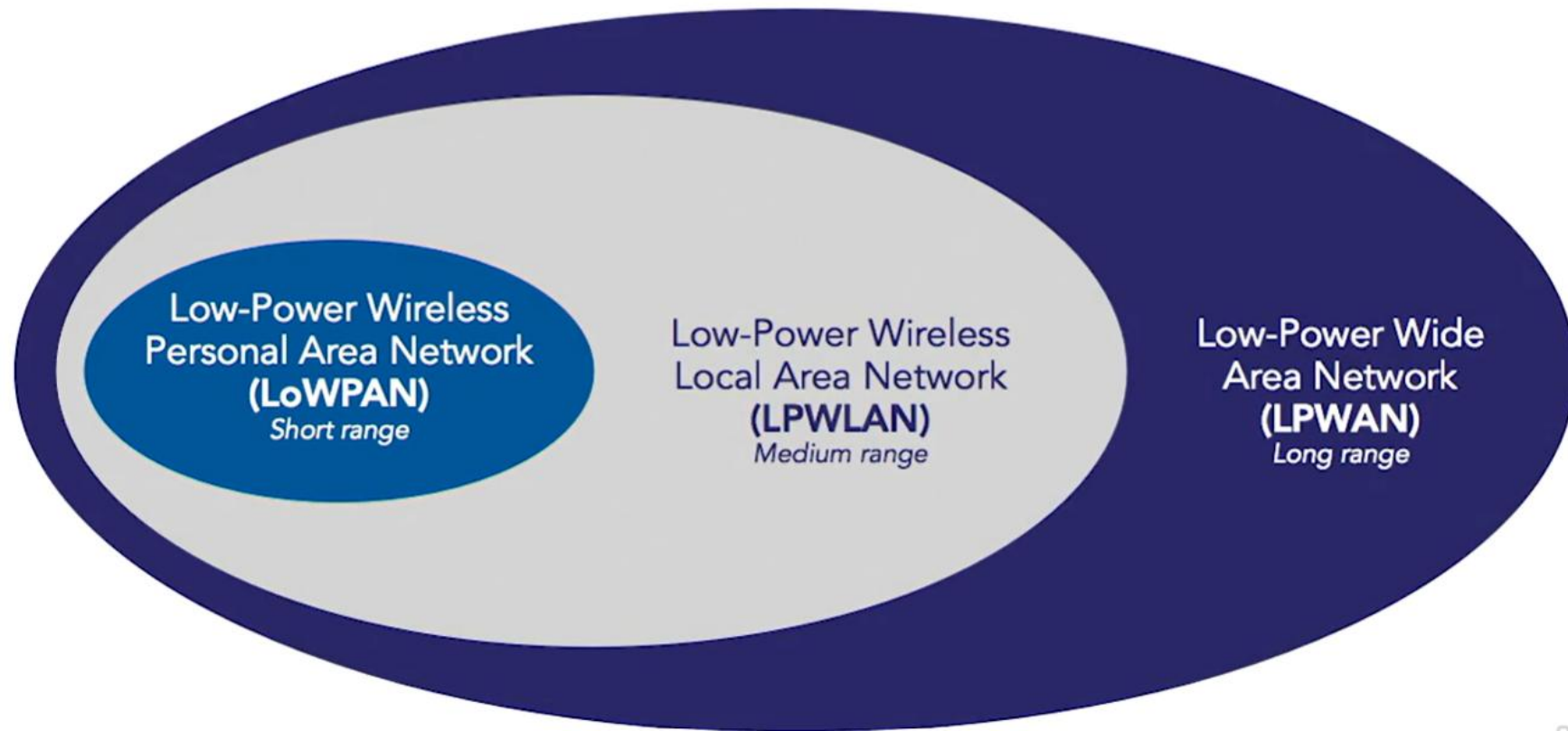
- Machine condition monitoring
- Process Automation
- Asset management
- And more



Advantage of wireless networks

- ▶ 1. Flexible location of the wireless devices
- ▶ 2. Effortless installation
- ▶ 3. Network scalability
- ▶ 4. Flexible network capacity

Wireless networks range



Wireless personal Area Network

- ▶ A wireless network consisting of a set of wireless devices within a person's workspace to exchange and share data.

LoWPAN Features

- ▶ Low-power, low cost devices
- ▶ Low power radio modules and protocol
- ▶ Usually low rate networks
- ▶ Low-power IoT application

Examples of LoWPAN Technologies

- ▶ Bluetooth Low Energy
- ▶ ZigBee
- ▶ 6LoWPAN
- ▶ WirelessHART

Wireless Wide Area Networks

- ▶ A wireless network with a coverage of a large area, such as a metropolitan area.

Example of WWANs

- ▶ GSM-based mobile networks (Global System for Mobile communication)
- ▶ UMTS-based mobile networks (Universal mobile telecommunication system, 3g)
- ▶ LTE-advanced based mobile networks

What is a LPWAN?

- ▶ It is suitable for machine to machine (M2M) communications.
- ▶ Low-rate machine or sensing data are required
- ▶ Frequent battery replacement or a constant power source is unavailable.
- ▶ A high throughput and/Or high performance may not be required.

LPWAN Features

- ▶ Massive number of devices
- ▶ Easy and low-cost deployment
- ▶ Low-data usage
- ▶ Long battery life.
- ▶ Low end application

Example of LPWAN Technologies

- ▶ LTE-M(long-term evolution for machine type communications)
- ▶ NB-IoT (narrowband IoT)
- ▶ EC-GSM-IoT(extended coverage GSM IoT)
- ▶ LoRa
- ▶ SIGFOX

Challenge

- ▶ Check the available wireless networking technologies on your smart phone
- ▶ Identify the types of these wireless networking technologies

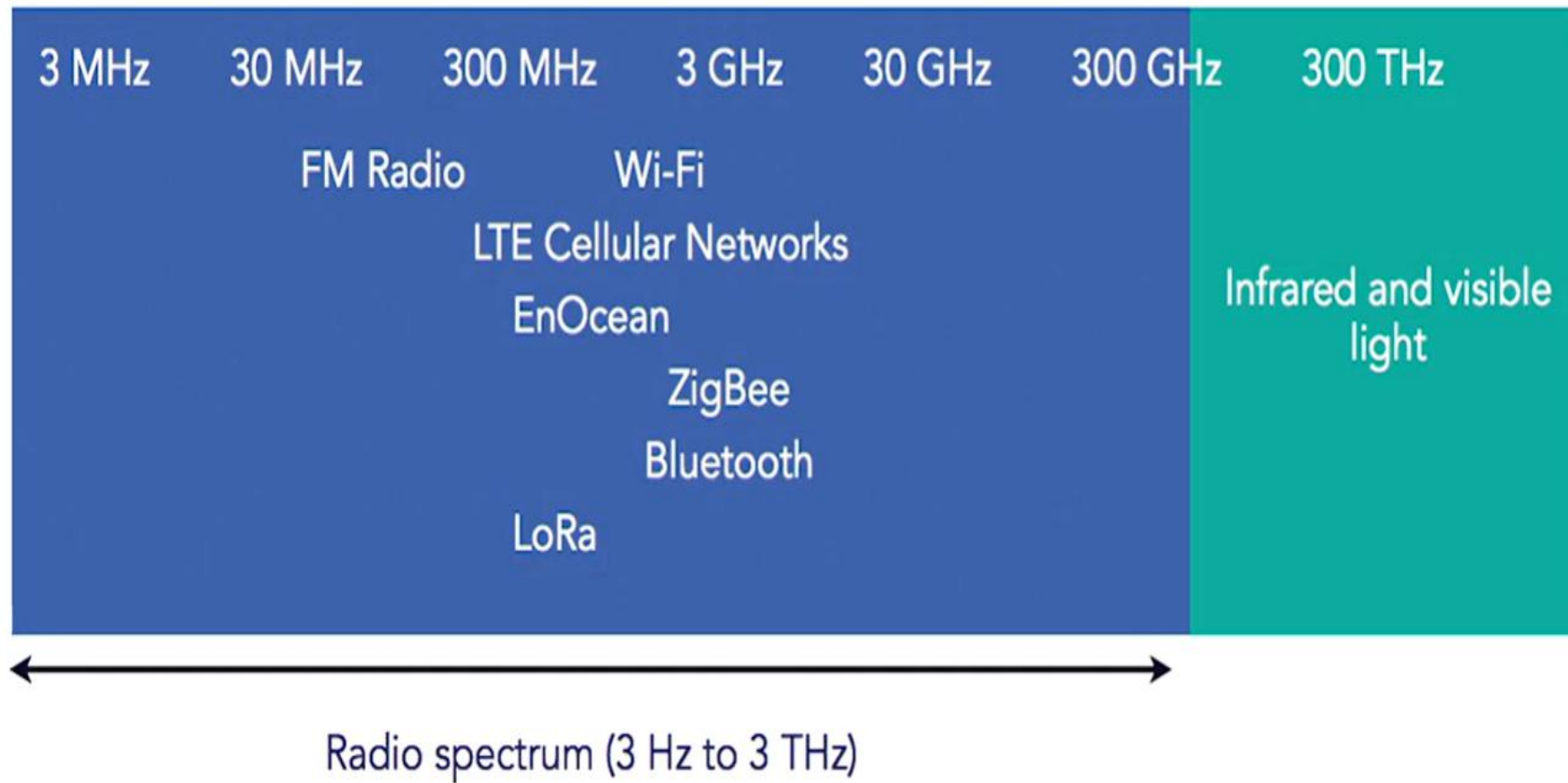
Wireless networking technology

Software Update	>
AirDrop	>
AirPlay & Handoff	>
Picture in Picture	>
CarPlay	>
Home Button	>
iPhone Storage	>
Background App Refresh	>

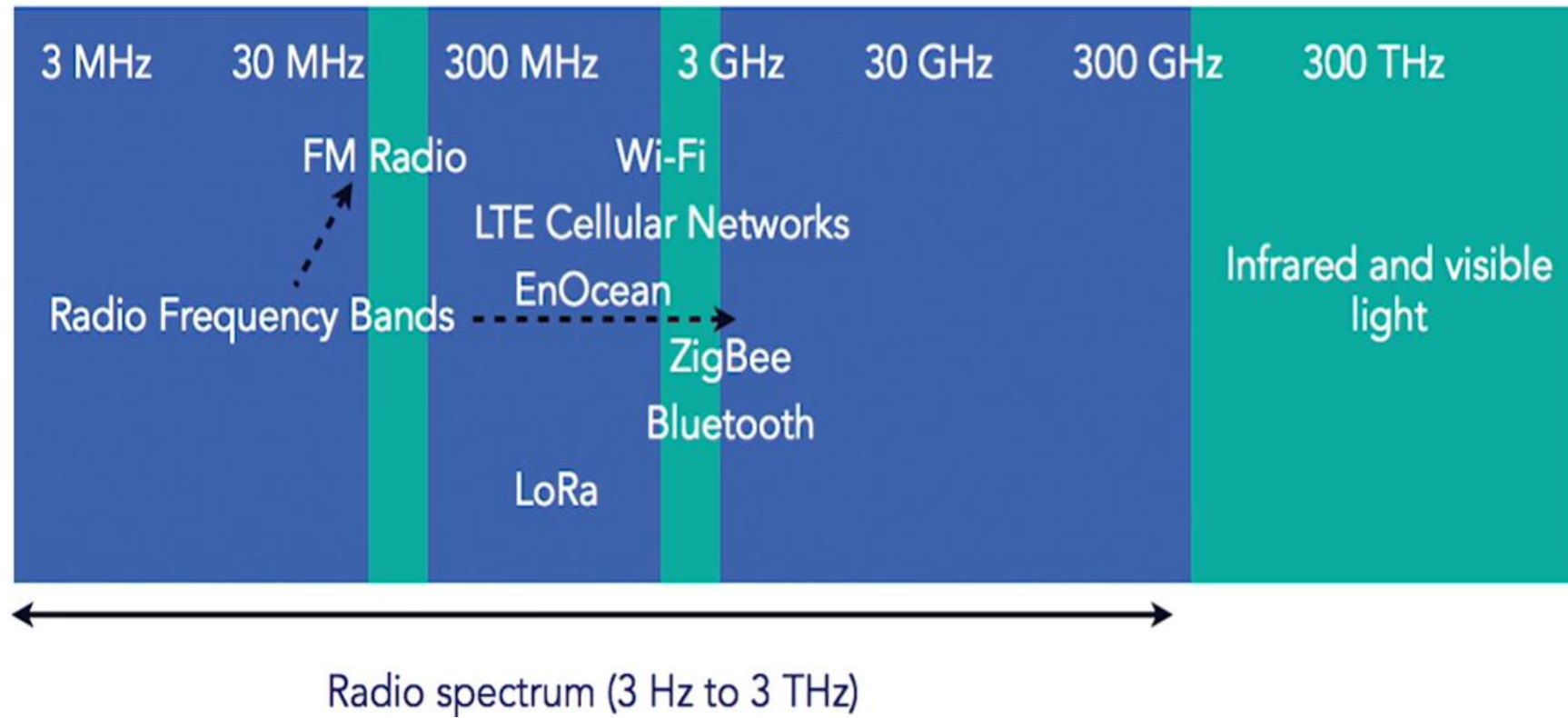
Bluetooth	Not Connected	>
Cellular		>
Personal Hotspot	Off	>
VPN	☐	
Notifications		>
Sounds & Haptics		>
Do Not Disturb		>
Screen Time		>
General		>

Capacity	128 GB
Available	64 GB
Carrier	
Wi-Fi Address	40:9C:28:03:02:01
Bluetooth	40:9C:28:03:02:02
IMEI	35 308510 583111 3
ICCID	89310800102574192863
Modem Firmware	4.02.01
SEID	>
Carrier Lock	No SIM restrictions

Radio spectrum



Radio spectrum



Radio Frequency Bands

- ▶ Unlicensed frequency bands

Wifi and Bluetooth: 2.4GHz and 5 GHz frequency bands

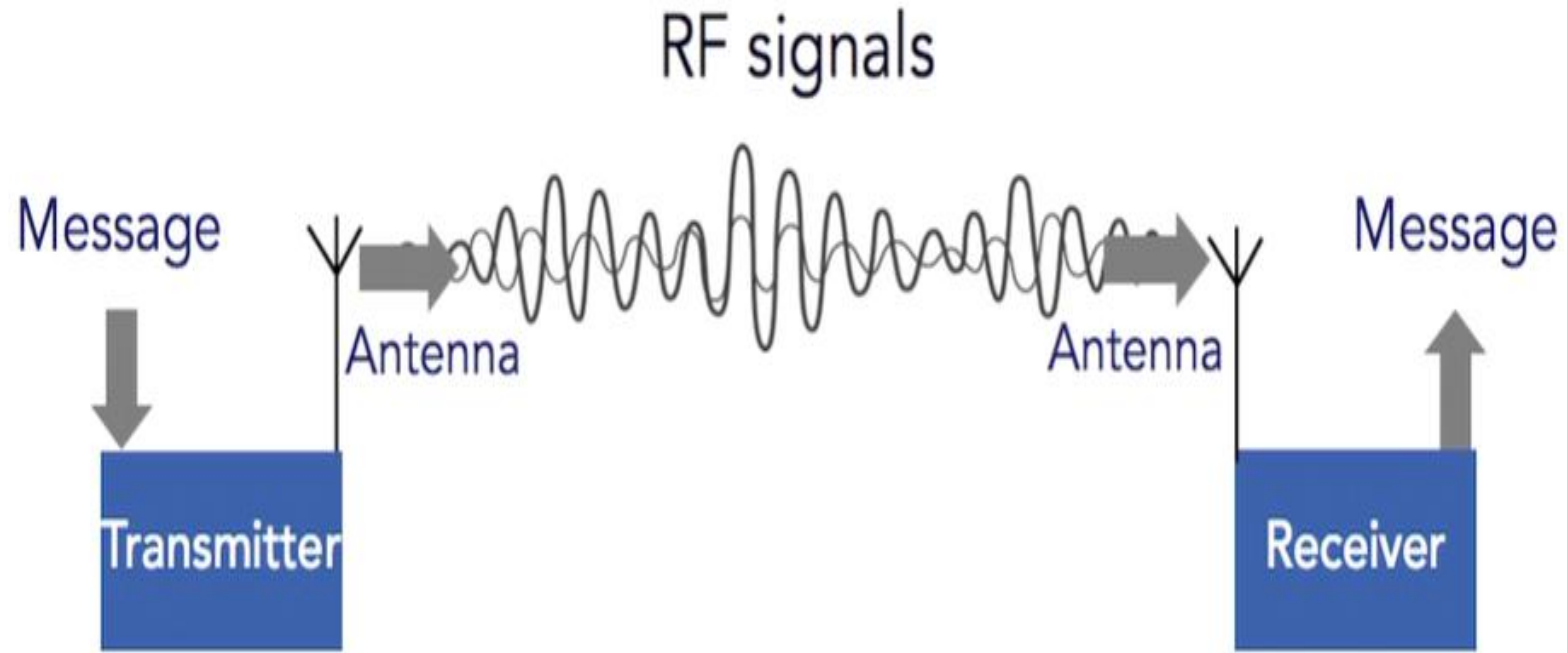
LoRa and EnOcean: sub-1 GHz frequency bands

Licensed Frequency bands

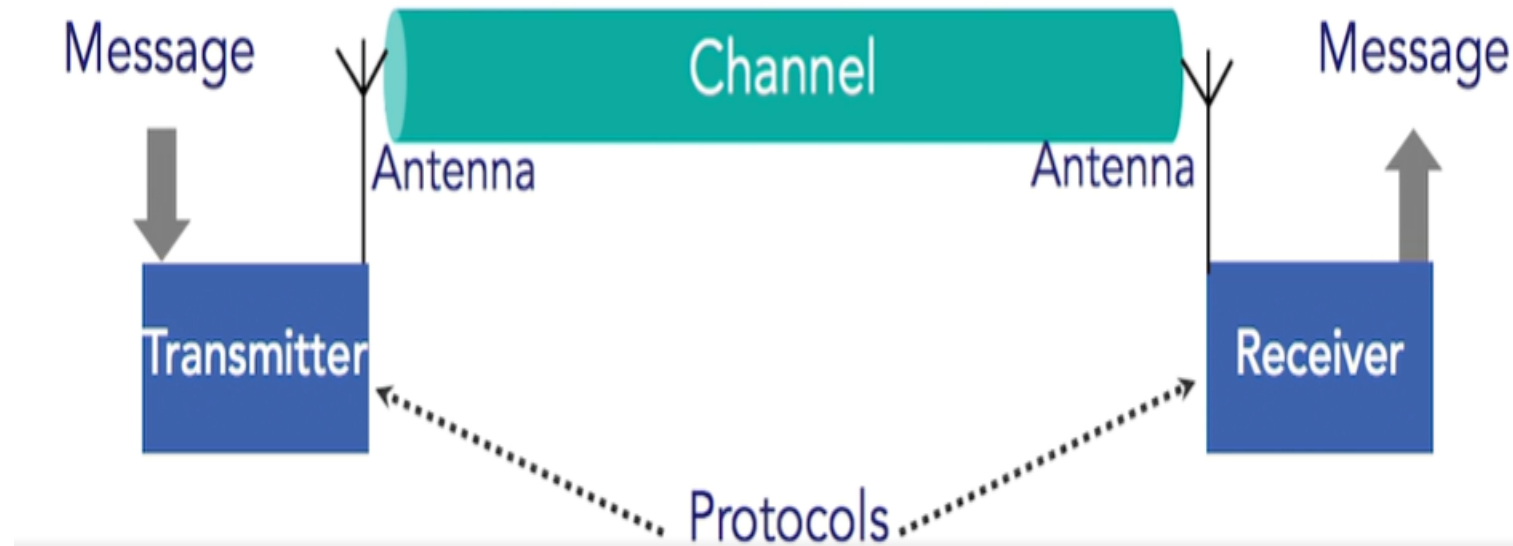
LTE cellular networks: various frequency band option



A wireless communication system



A wireless communication system

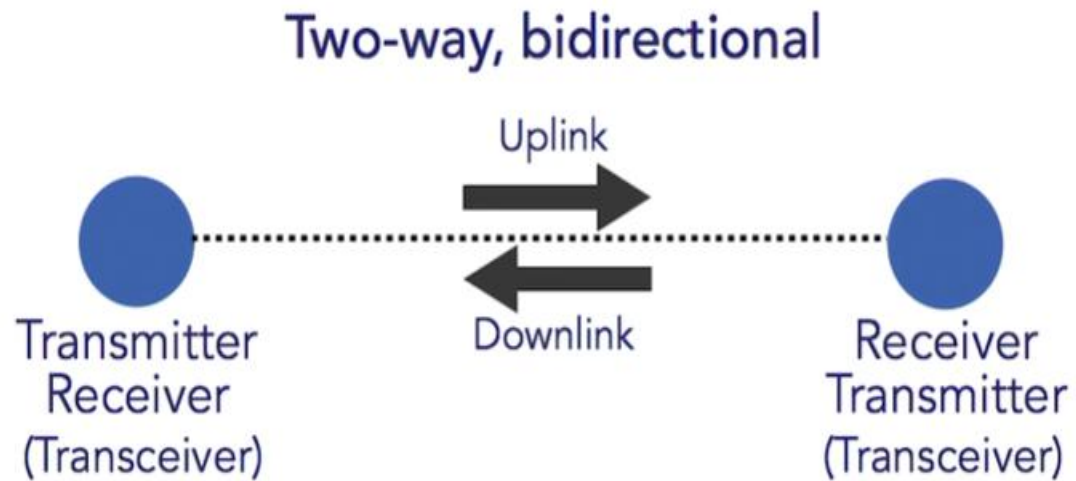


Types of communication

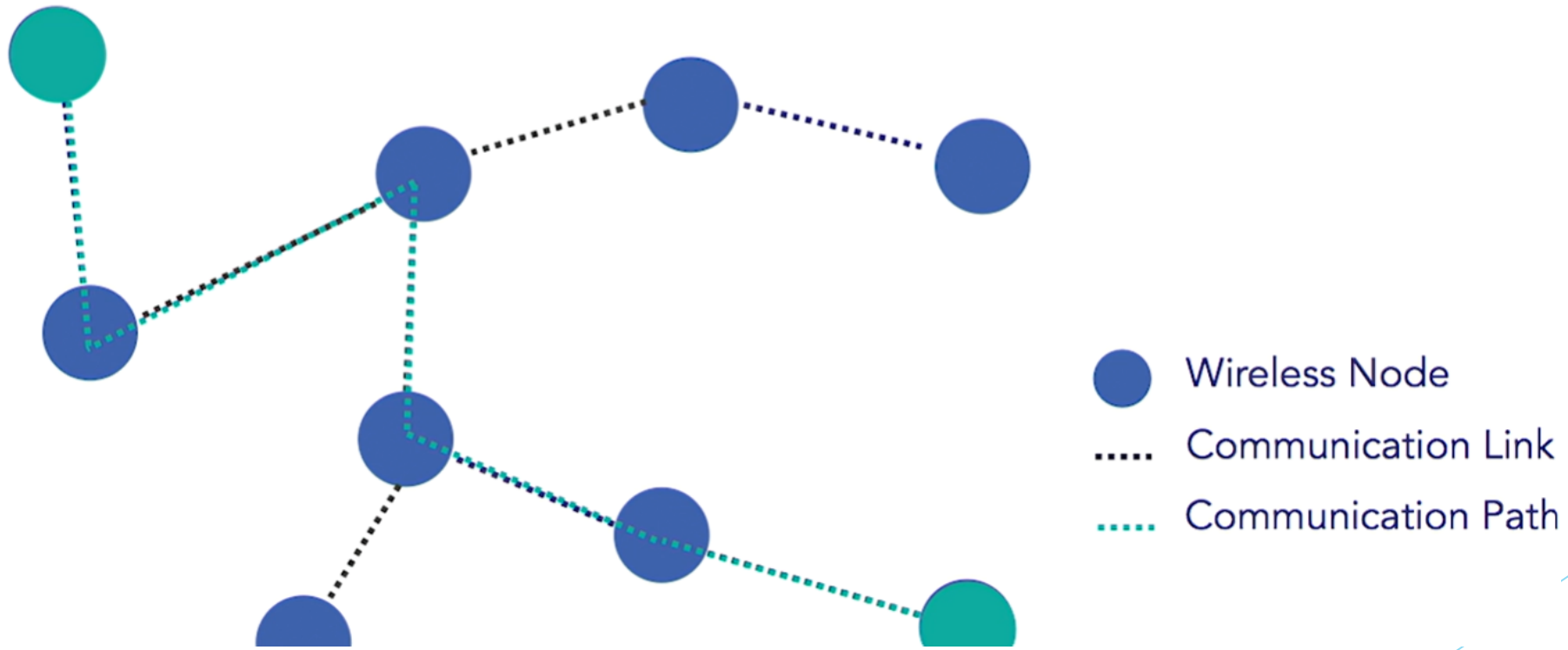
One-way, unidirectional



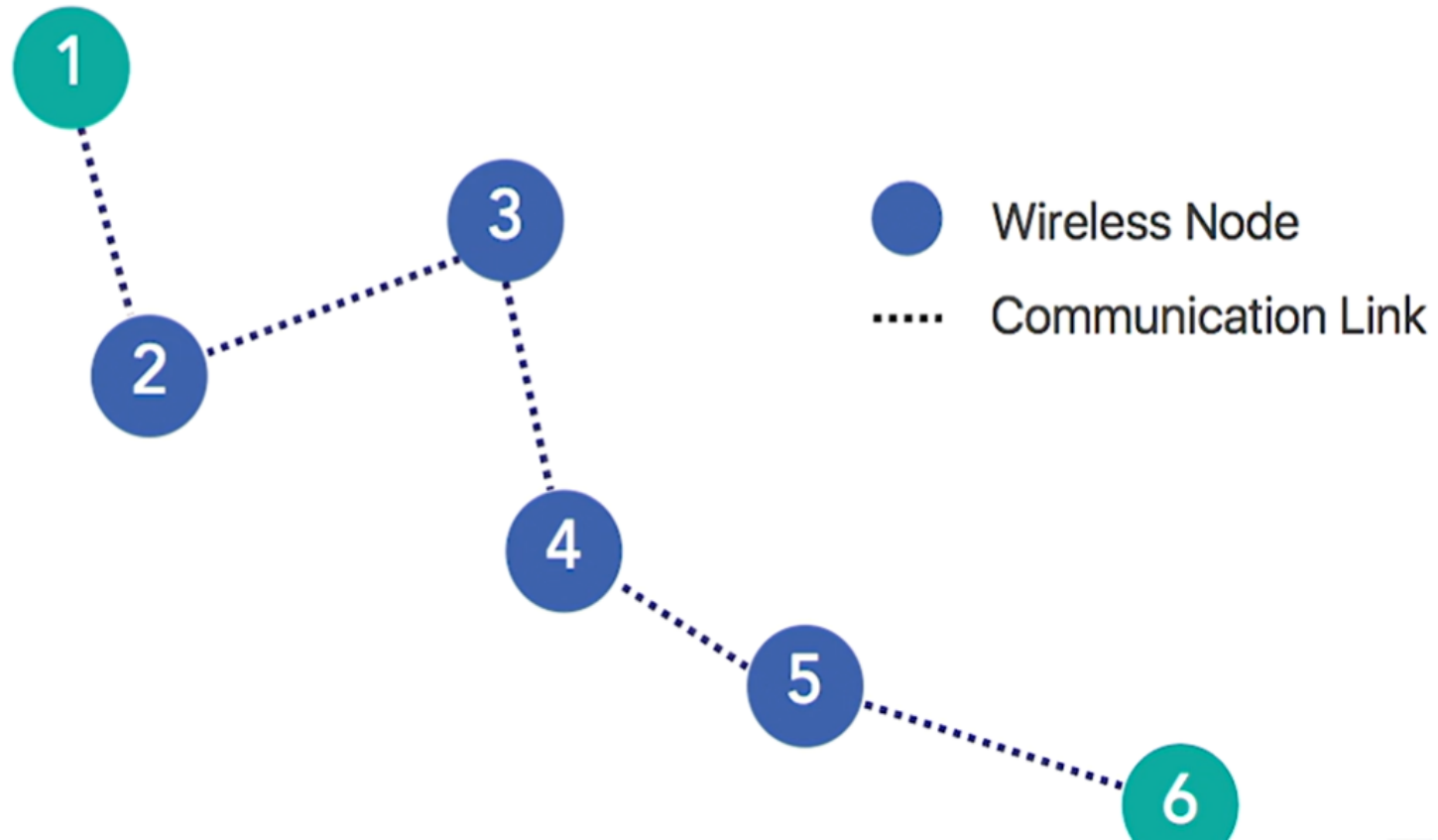
Types of communication



A wireless network



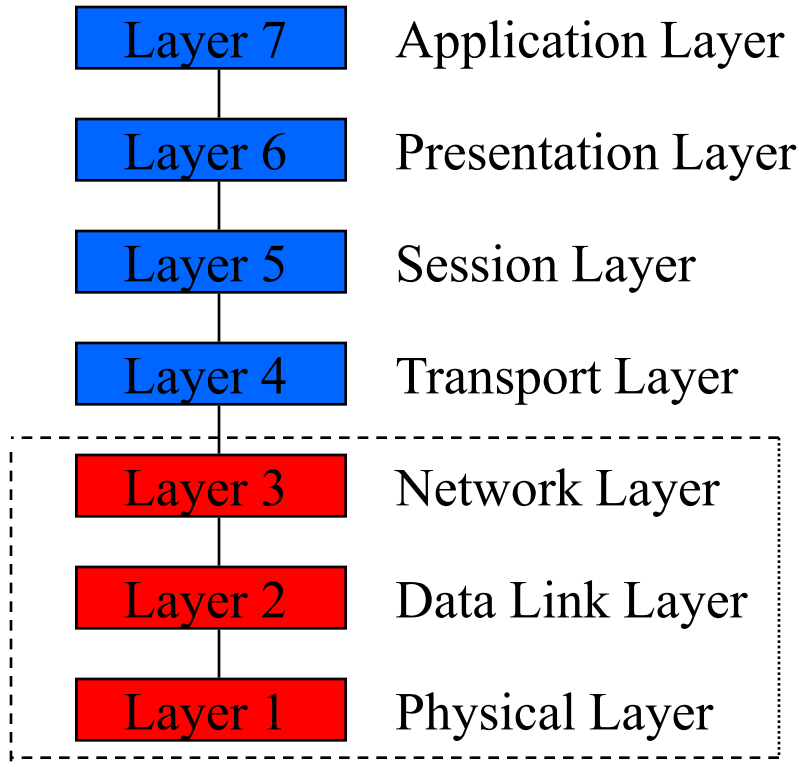
OSI reference model



OSI Reference Model

- ▶ OSI Reference Model - internationally standardised network architecture.
- ▶ OSI = *Open Systems Interconnection*: deals with *open systems*, i.e. systems open for communications with other systems.
- ▶ Specified in ISO 7498.
- ▶ Model has 7 layers.

7-Layer OSI Model



- ▶ Layers 1-4 relate to communications technology.
- ▶ Layers 5-7 relate to user applications.

Communications subnet boundary

Layer 7: Application Layer

- ▶ Level at which applications access network services.
- ▶ Represents services that directly support software applications for file transfers, database access, and electronic mail etc.

Layer 6: Presentation Layer

- ▶ Related to representation of transmitted data
 - ▶ Translates different data representations from the Application layer into uniform standard format
- ▶ Providing services for secure efficient data transmission
 - ▶ e.g. data encryption, and data compression.

Layer 5: Session Layer

- ▶ Allows two applications on different computers to establish, use, and end a session.
 - ▶ e.g. file transfer, remote login
- ▶ Establishes dialog control
 - ▶ Regulates which side transmits, plus when and how long it transmits.
- ▶ Performs *token management* and *synchronization*.

Layer 4: Transport Layer

- ▶ Manages transmission packets
 - ▶ Repackages long messages when necessary into small packets for transmission
 - ▶ Reassembles packets in correct order to get the original message.
- ▶ Handles error recognition and recovery.
 - ▶ Transport layer at receiving acknowledges packet delivery.
 - ▶ Resends missing packets

Layer 3: Network Layer

- ▶ Manages addressing/routing of data within the subnet
 - ▶ Addresses messages and translates logical addresses and names into physical addresses.
 - ▶ Determines the route from the source to the destination computer
 - ▶ Manages traffic problems, such as switching, routing, and controlling the congestion of data packets.
- ▶ Routing can be:
 - ▶ Based on static tables
 - ▶ determined at start of each session
 - ▶ Individually determined for each packet, reflecting the current network load.

Layer 2: Data Link Layer

- Packages raw bits from the Physical layer into frames (logical, structured packets for data).
- Provides reliable transmission of frames
 - It waits for an acknowledgment from the receiving computer.
 - Retransmits frames for which acknowledgement not received

Layer 1: Physical Layer

- ▶ Transmits bits from one computer to another
- ▶ Regulates the transmission of a stream of bits over a physical medium.
- ▶ Defines how the cable is attached to the network adapter and what transmission technique is used to send data over the cable. Deals with issues like
 - ▶ The definition of 0 and 1, e.g. how many volts represents a 1, and how long a bit lasts?
 - ▶ Whether the channel is simplex or duplex?
 - ▶ How many pins a connector has etc.

Internet Protocols vs OSI

Application		Application
Presentation		
Session		
Transport		TCP
Network		IP
Data Link		Network Interface
Physical		Hardware

- ▶ Explicit Presentation and session layers missing in Internet Protocols
- ▶ Data Link and Network Layers redesigned

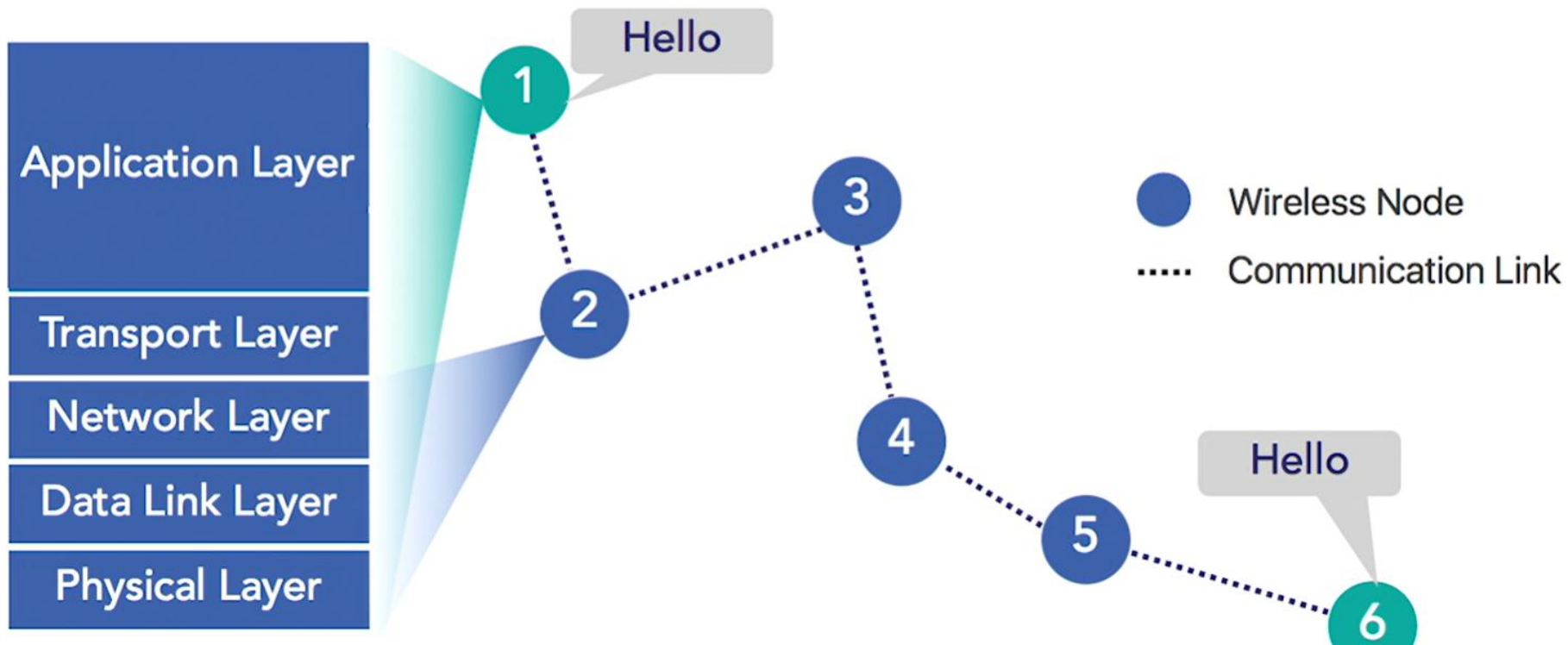
Services in the OSI Model

- ▶ In OSI model, each layer provide services to layer above, and ‘consumes’ services provided by layer below.
- ▶ Active elements in a layer called *entities*.
- ▶ Entities in same layer in different machines called *peer entities*.

Connections

- ▶ Layers can offer *connection-oriented* or *connectionless* services.
- ▶ Connection-oriented like telephone system.
- ▶ Connectionless like postal system.
- ▶ Each service has an associated *Quality-of-service* (e.g. reliable or unreliable).

OSI reference model



Protocol Stack

- A set of communications protocols
- Often implemented in a software stack, such as a TCP/IP, ZigBee, 6LoWPAN, or Bluetooth stack

Some common protocols

- Internet protocols (IPv4 and IPv6)
- Routing protocols
- Security protocols
- Error reporting and diagnostic protocols

IoT device hardware

Microcontroller

Radio module

Power supply

Sensors/actuators

Other components

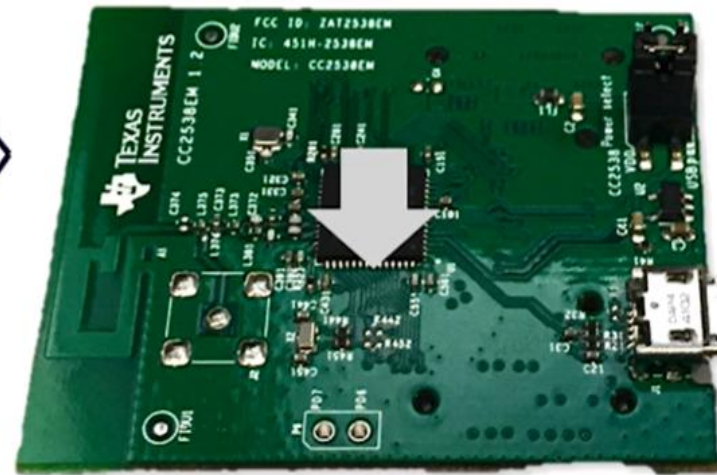
Sensors



Air Quality



Temperature &
Humidity



Power consumption on an IoT device

Hardware components

Radio module, microcontroller, sensors/actuators, LEDs, etc.

Communication software

Power consumption on an IoT device

System and application software

Operating system

Message encryption algorithms, middleware, cloud connector software, etc.

Power consumption on an IoT network

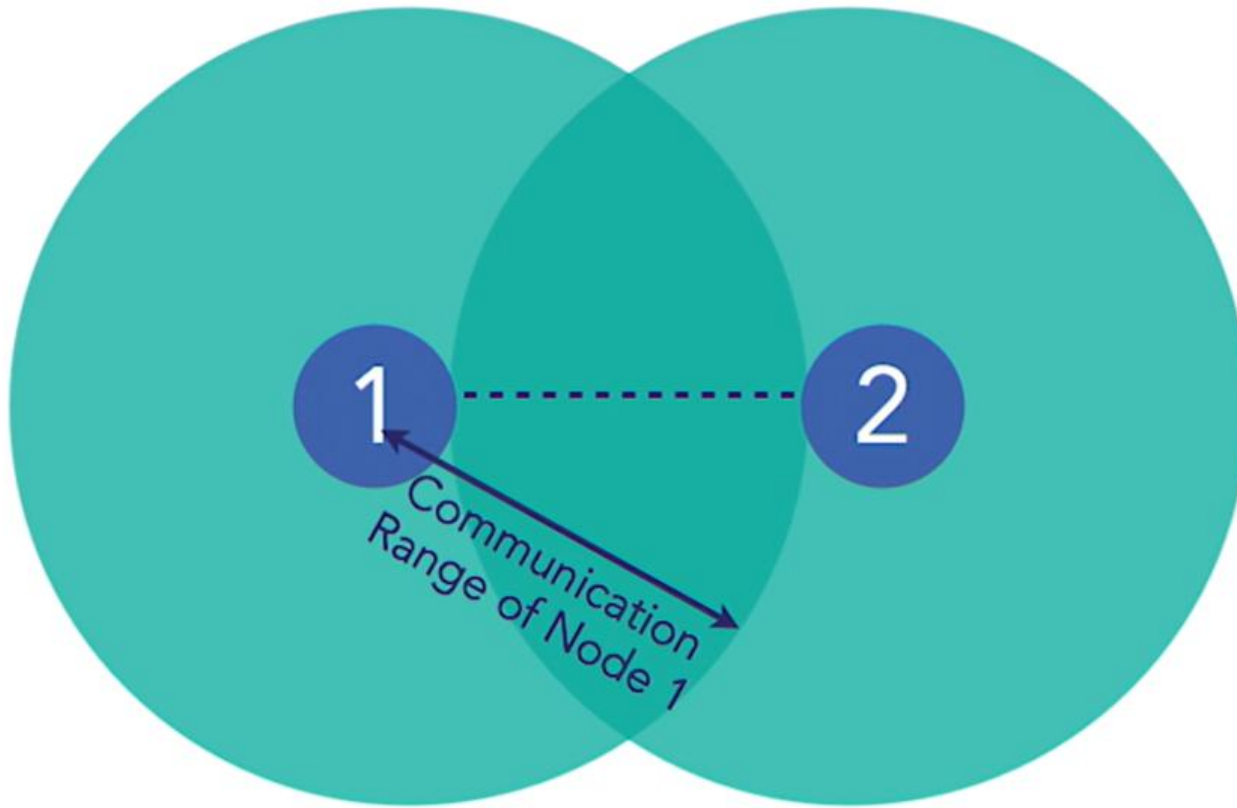
End devices (usually battery powered or battery-less)

Sensor nodes, etc.

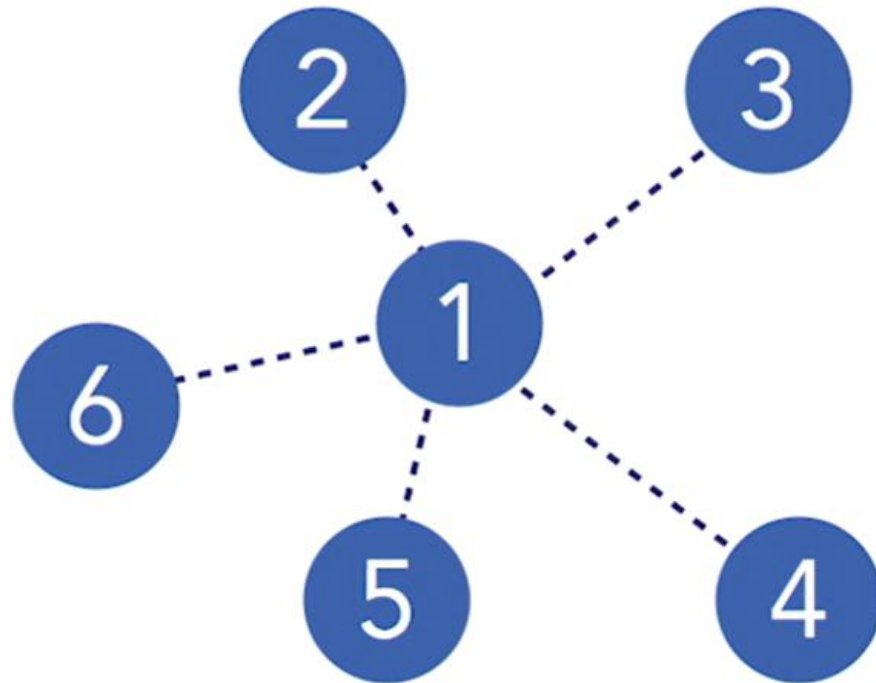
Backbone devices (usually battery or mains powered)

Gateways, base stations, routers, etc.

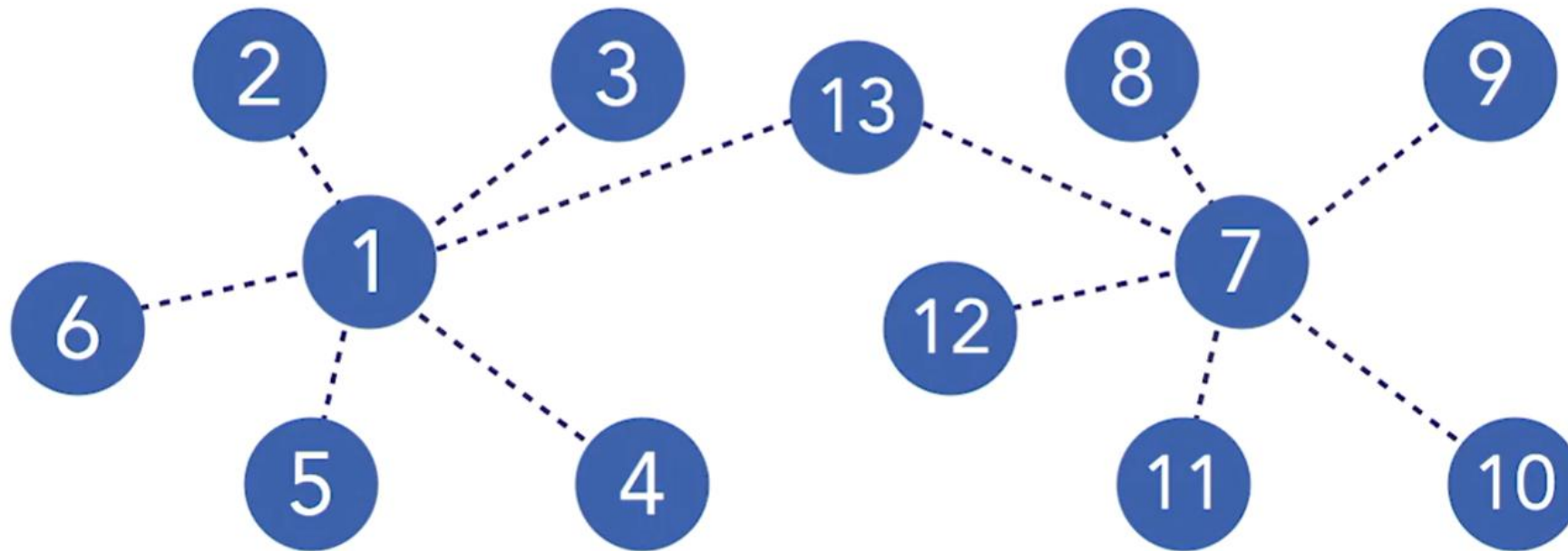
Point to point connection



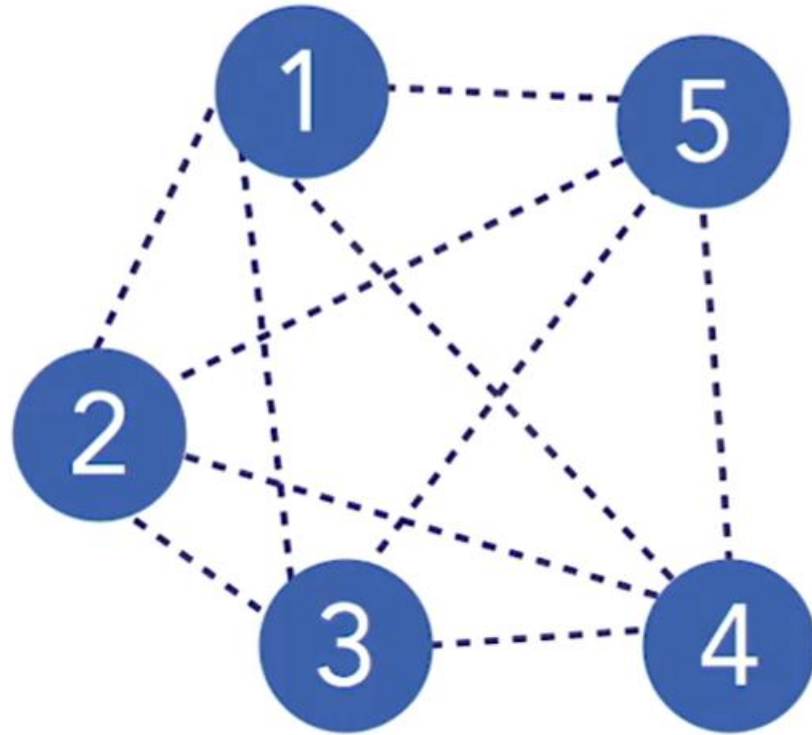
Star Topology



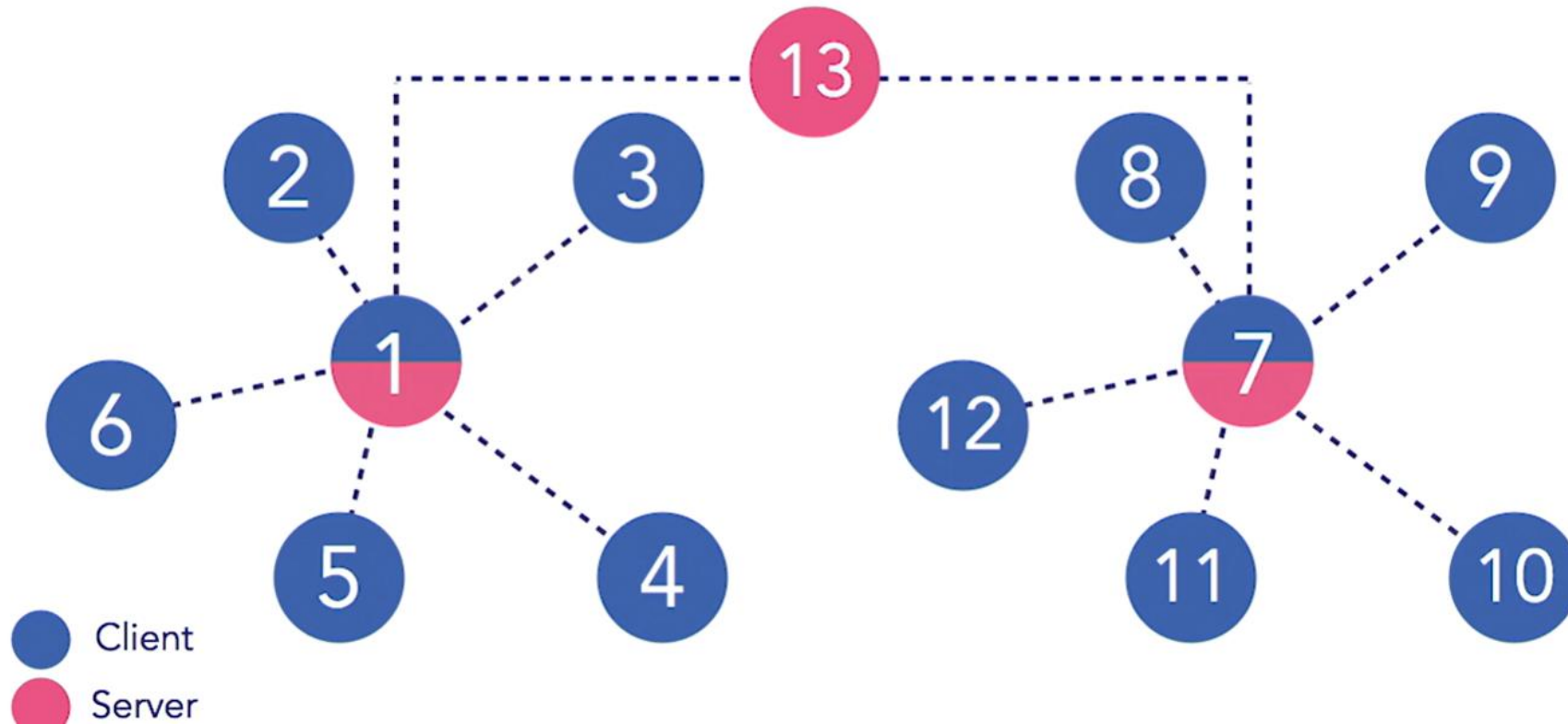
Extended star topology



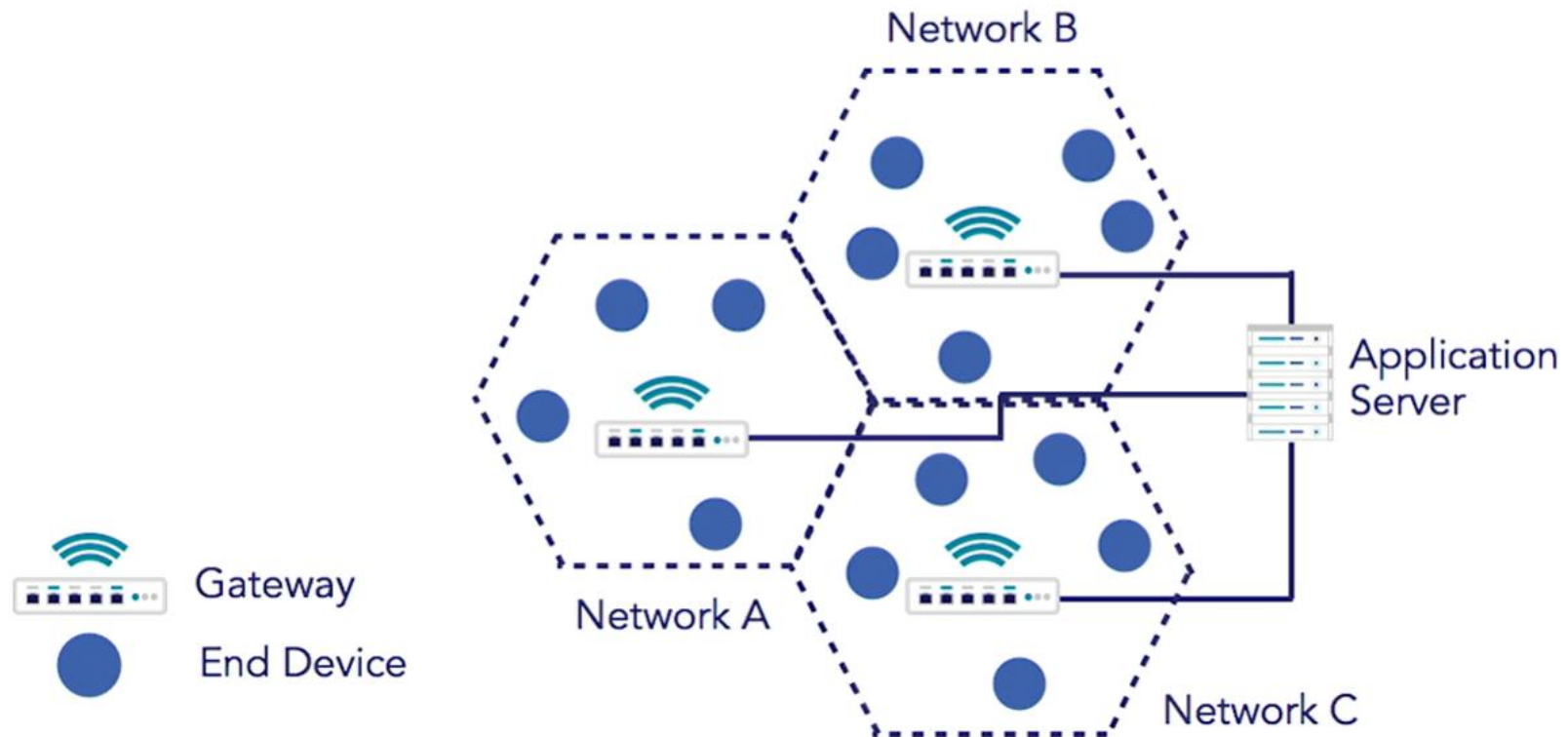
Mesh Topology



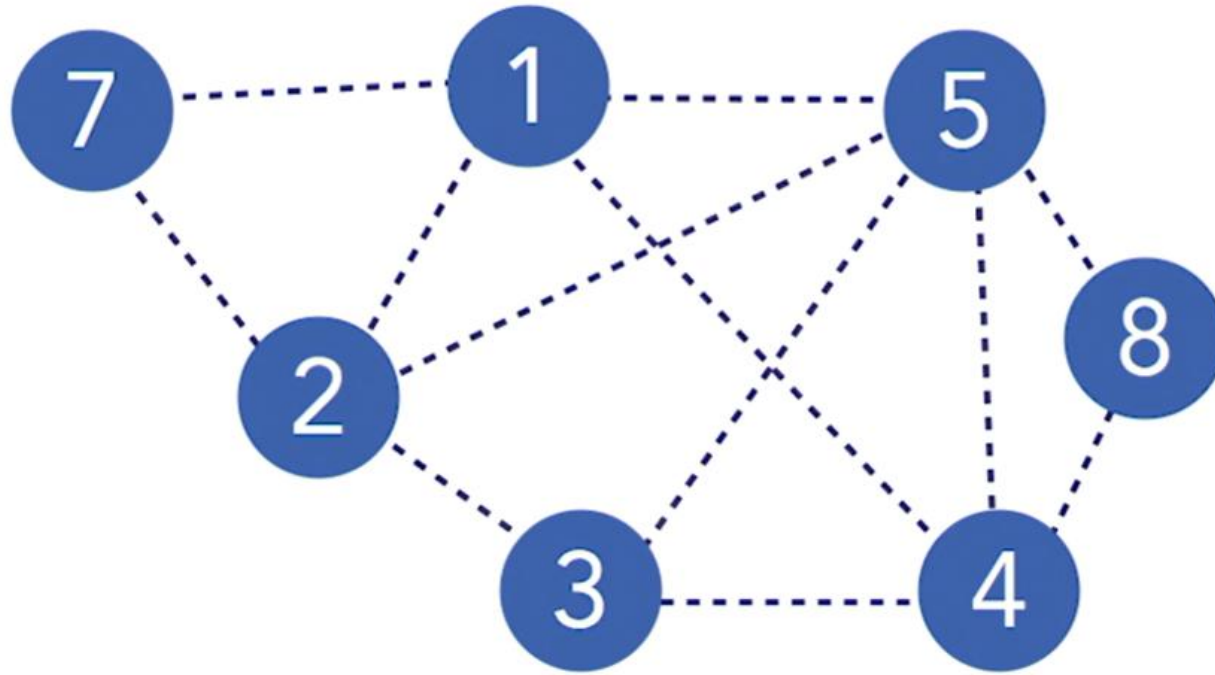
Client server architecture



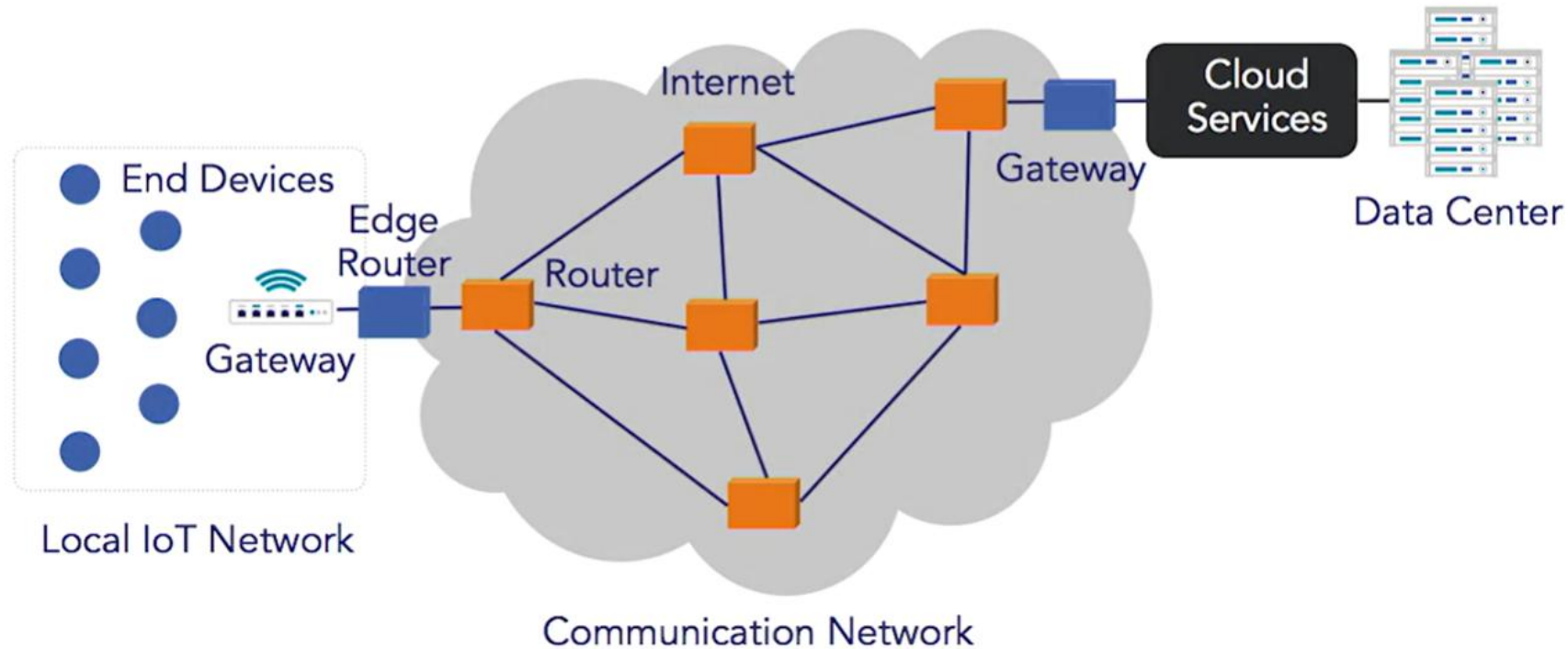
Example of Client server architecture



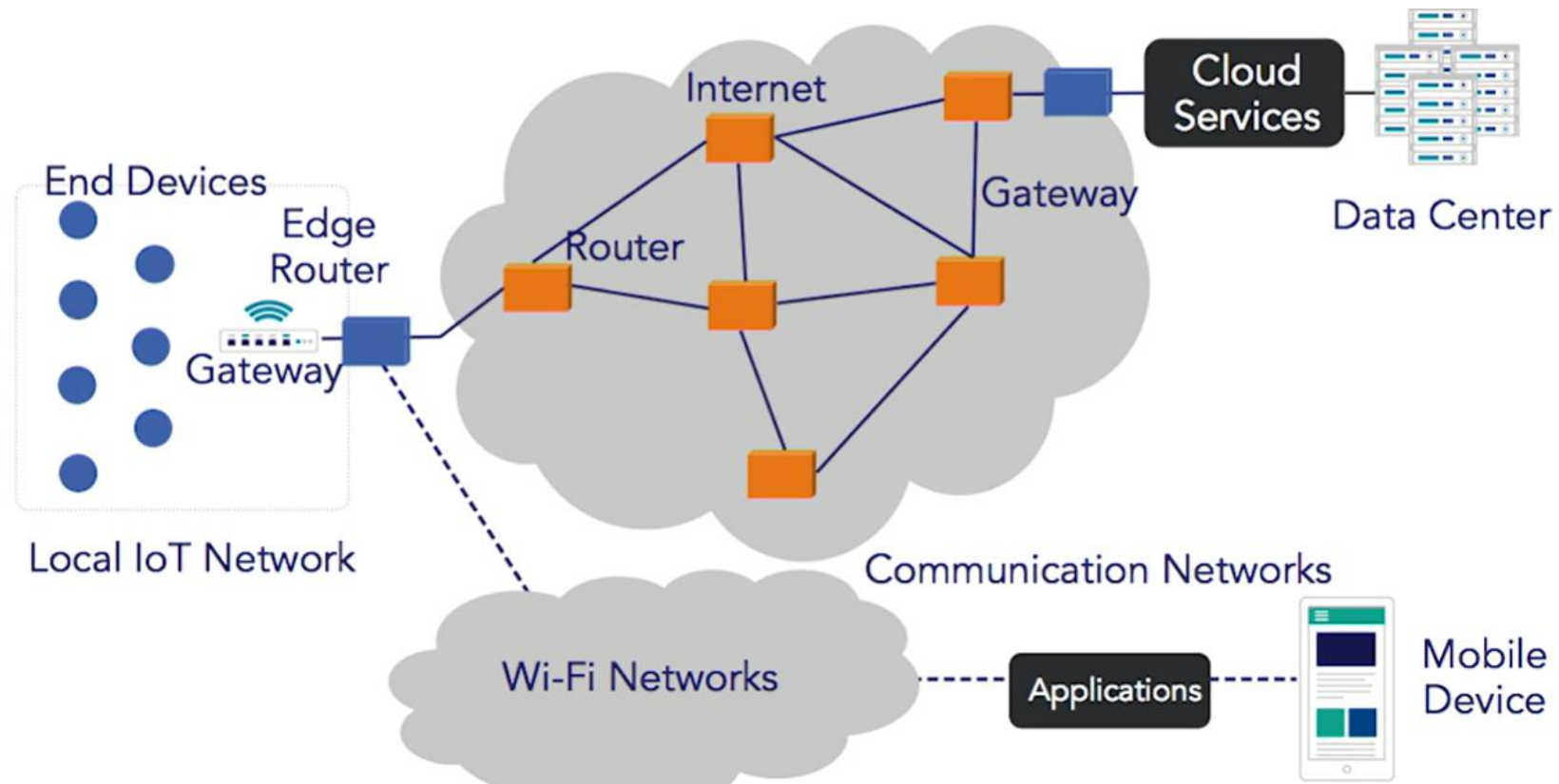
Peer to peer Architecture



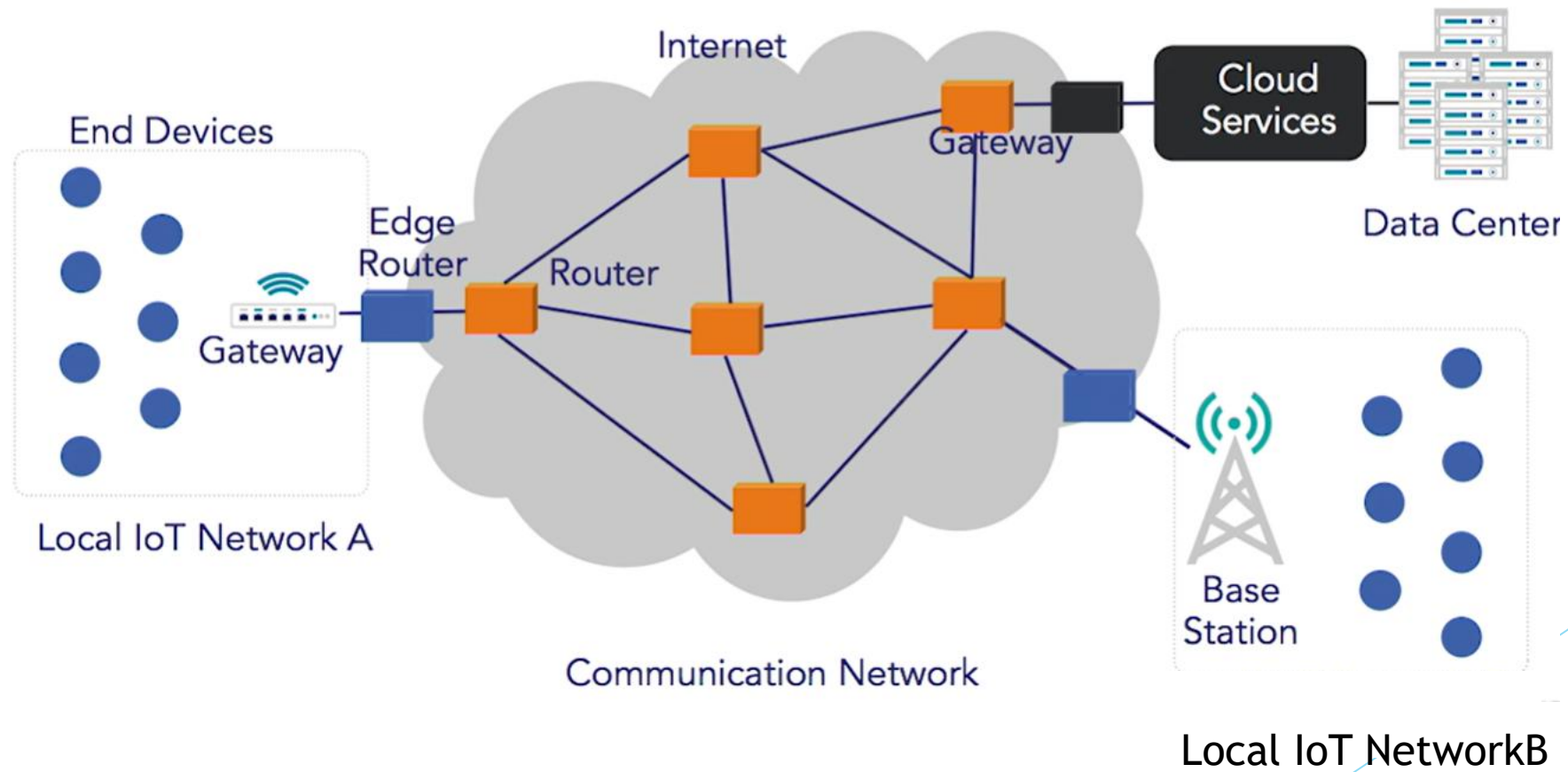
Point to point connection



Point to point connection



Point to point connection



Key Elements

- ▶ End device:

A device with the mandatory capabilities of communication and optional capabilities of sensing, actuation, data storage, and data processing.

- ▶ Gateway:

Software stack for radio communication and networking.

- ▶ Edge router:

A network device located at the boundary of a network connecting external communication network.

Key Elements

- ▶ Base station

A radio transceiver connecting a number of end devices , for example the base station for cellular network

- ▶ Communication network:

A backbone network , such as internet or Wifi network.

- ▶ Cloud service

A cloud computing service on the internet , e.g. SAAS application for data storage, visualization analysis and IoT device management.

IoT security issues

Confidentiality

Protect the data from unauthorized disclosure

Integrity

Make sure data are not tampered

Availability

Ensure the services are available

References

- ▶ 1. IoT - Low power wireless networking

Question



Thank you

